IoT-Based Real-Time Nutrient Stress Detection using Hybrid CNN-LSTM Model for Smart Agriculture

Dheeraj M.G, Annepu Adarsh, Ch Rambhadra Kumar, Ch Punyamurthy
Indian Institute of Information Technology, Vadodara
Guide: Dr. Kamal Kishor Jha | Course: Embedded System Project

***Abstract*—**Agricultural productivity is fundamentally dependent on soil nutrient availability and environmental conditions. Traditional soil testing methods are time-consuming, costly, and unsuitable for real-time field monitoring. This paper proposes a low-cost embedded IoT system integrated with a hybrid deep learning model for automated real-time nutrient stress detection. The hardware layer uses an Arduino Uno microcontroller interfaced with a soil moisture sensor, DHT11 temperature and humidity sensor, LDR light intensity sensor, and a pH sensor module. The machine learning layer employs a parallel hybrid CNN-LSTM architecture where the Convolutional Neural Network (CNN) branch captures spatial feature interactions among Nitrogen (N), Phosphorus (P), Potassium (K), and pH inputs, while the Long Short-Term Memory (LSTM) branch models sequential temporal dependencies. The dataset is derived from Government of India soil data cards and labeled using threshold-based heuristics into three classes: Normal, Deficient, and Stress. Class imbalance is handled using computed class weights during training. The model is trained with Adam optimizer, categorical cross-entropy loss, EarlyStopping callback (patience=5), and Dropout regularization over up to 50 epochs, achieving a test accuracy of approximately 96–99%. The system provides a cost-effective solution for precision agriculture, enabling real-time soil health monitoring without laboratory testing.

Key Terms*—CNN-LSTM, class weights, deep learning, dropout, early stopping, embedded systems, IoT, nutrient stress detection, precision agriculture, soil classification.*

--Punyamurthy

I. INTRODUCTION - Punyamurthy

AGRICULTURE remains the backbone of global food security and economic stability. With the world population projected to approach 10 billion by 2050, food production must increase significantly. Arable land per capita is simultaneously decreasing while climate variability is intensifying, making efficient nutrient management a critical determinant of sustainable crop yield.

Soil macronutrients—Nitrogen (N), Phosphorus (P), and Potassium (K)—alongside soil pH directly influence plant growth, photosynthesis, and disease resistance. A deficiency or imbalance in any of these nutrients causes nutrient stress, manifesting as stunted growth, chlorosis, or premature leaf drop, ultimately reducing agricultural output.

Conventional soil testing requires laboratory analysis, which is expensive and time-consuming. Farmers often rely on assumptions, leading to improper fertilizer usage and reduced yield. This is particularly difficult for small and marginal farmers who constitute the majority of the agricultural workforce in India.

The convergence of the Internet of Things (IoT) and Artificial Intelligence (AI) has opened new avenues for smart agriculture. IoT sensors collect real-time environmental data continuously at negligible cost. Machine learning models analyze these data streams to identify patterns and make predictions. When combined in an embedded system, they enable on-the-fly nutrient stress detection without external laboratory infrastructure [8].

This work presents an end-to-end smart agriculture system consisting of: (i) an Arduino Uno sensor node collecting environmental data; (ii) threshold-based embedded logic for rapid initial classification; and (iii) a parallel hybrid CNN-LSTM model with Dropout regularization, EarlyStopping, and class weight balancing, trained on Government of India soil data for three-class prediction—Normal, Deficient, and Stress.

II. RELATED WORK - Adarsh

The integration of IoT and machine learning for agricultural monitoring has gained significant momentum. This section reviews prior work across three areas directly relevant to this project.

A. IoT-Based Soil Monitoring

Wireless sensor network based systems have been proposed for monitoring soil nutrients and pH in greenhouses using low-cost sensors for real-time data logging [2]. However, such systems lacked intelligent decision-making, relying instead on static threshold comparisons. Other IoT platforms demonstrated continuous field monitoring of soil NPK and moisture [4], but inference remained limited to rule-based logic without deep feature learning.

B. Machine Learning for Soil Classification

Classical machine learning models including Decision Trees, Random Forest, and Support Vector Machines have been applied to soil nutrient classification with accuracies in the 88–92% range [9]. These models fail to capture complex non-linear feature interactions in multi-sensor soil data. Reviews of machine learning in agriculture confirm deep learning models are consistently superior when sufficient labeled data is available [9].

C. Hybrid Deep Learning for Agricultural Data

CNN-based architectures demonstrate effectiveness in hierarchical spatial feature extraction, while LSTM networks enable learning of long-range temporal dependencies crucial for sensor time-series data [5], [6]. A Parallel Hybrid CNN-LSTM model using IoT sensors for real-time macronutrient assessment achieved approximately 98% accuracy [1]. A hybrid CNN-LSTM model for diagnosing rice nutrient levels demonstrated that combining CNN spatial extraction with LSTM temporal modeling produces more

robust nutrient deficiency detection than either architecture alone [11]. Surveys of deep learning in agriculture confirm CNN-LSTM hybrid methods are the leading approaches for time-series sensor data in precision agriculture [7].

III. PROPOSED SYSTEM - Dheeraj MG

The proposed system consists of two main components: a hardware sensing layer and a software machine learning layer, both designed to be low-cost and deployable in resource-constrained agricultural settings.

A. System Architecture

Fig. 1 illustrates the proposed system block diagram. The complete workflow is:

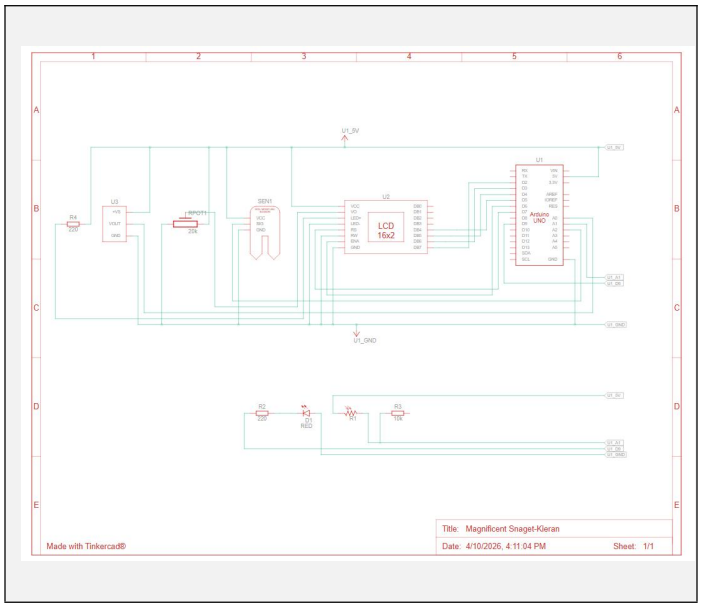

Fig. 1. Proposed System Architecture Block Diagram.

1. Sensors collect analog signals via Arduino Uno (soil moisture, temperature, humidity, light intensity).
2. Arduino applies threshold-based initial stress detection in embedded firmware.
3. Soil dataset (N, P, K, pH) is loaded, preprocessed, and labeled for ML training.
4. CNN-LSTM model is trained with class weights and EarlyStopping.
5. Model outputs one of three labels: Normal, Deficient, or Stress.
6. Result displayed on LCD with color-coded LED indication.

B. Hardware Layer

The hardware layer is built around the Arduino Uno microcontroller (ATmega328P, 16 MHz, 32 KB Flash). The following sensors are interfaced as shown in Fig. 2:

- Soil Moisture Sensor (Resistive): analog output 0–1023 mapped to 0–100% moisture.
- DHT11: ambient temperature (0–50°C, ±2°C) and relative humidity (20–90% RH).
- LDR (Light Dependent Resistor): light intensity via voltage divider.
- pH Sensor Module: soil pH (0–14 range) via analog voltage proportional to hydrogen ion concentration.

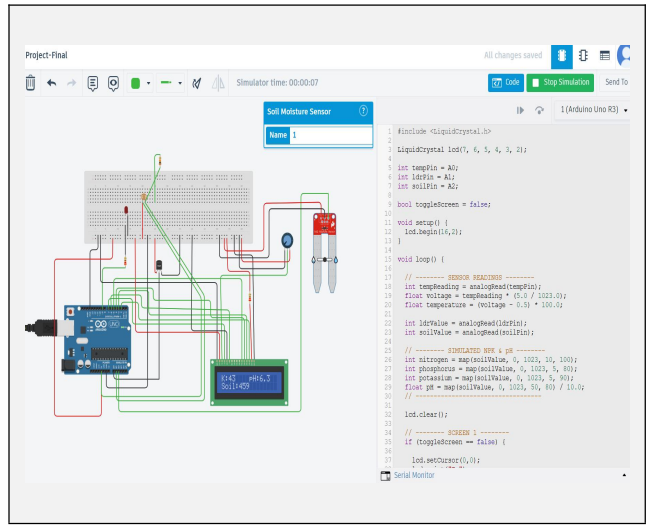

Fig. 2. Embedded System Circuit Design (Tinkercad Simulation).

Link - https://www.tinkercad.com/things/iZoJfhCpb4h-embedded-project/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits

C. Software — CNN-LSTM Model

The software layer is implemented in Python using TensorFlow/Keras. The parallel hybrid CNN-LSTM model takes normalized N, P, K, pH values as input and classifies soil condition into three categories. Fig. 3 shows the model architecture.

[We will add: Fig. 3 — CNN-LSTM Parallel Hybrid Architecture (Input (1,4) - CNN: Conv1D(32, kernel=1, relu) → Dropout(0.3) → Flatten → (32,) └─ LSTM: LSTM(32) → Dropout(0.3) → (32,) Concatenate → (64,) Dense(64, relu) → Dropout(0.4) Dense(3, softmax) → [Normal / Deficient / Stress])]

Fig. 3. Parallel Hybrid CNN-LSTM Model Architecture.

IV. DATASET AND PREPROCESSING - Rambhadra, Dheeraj

A. Data Source

The dataset is loaded from Government of India soil data cards hosted on Kaggle (dheerajmg/soil-data). The raw file contains district-wise records of macronutrient concentrations. After loading, the District column is dropped and the four remaining columns are renamed to N, P, K, and pH. Table I summarizes the dataset.

TABLE I Dataset Summary

Attribute	Detail
Source	Govt. of India Soil Data Cards (Kaggle: dheerajmg/soil-data)
Raw Features	District, N, P, K, pH
After Drop	N, P, K, pH (District removed)
Label Column	Stress (generated by threshold function)
Target Classes	Normal, Deficient, Stress
Normalization	StandardScaler (Z-score, fit on

Attribute	Detail
	train split)
Train / Test	80% / 20% (random_state = 42)

B. Label Generation — Threshold Function

The function stress_label() assigns a class to each row based on agronomic sufficiency thresholds for Indian soil. The logic used in the notebook is:

$$\text{Label} = \text{Deficient}, \quad \text{if } N<50 \text{ OR } P<30 \text{ OR } K<30 \quad (1)$$

$$\text{Label} = \text{Stress}, \quad \text{if } pH < 5.5 \text{ OR } pH > 8.0 \quad (2)$$

$$\text{Label} = \text{Normal}, \quad \text{otherwise} \quad (3)$$

Note: the pH lower threshold in this final version is 5.5 (updated from the earlier 5.0 used in preliminary work). Table II shows the resulting class distribution.

TABLE II Class Distribution After Threshold Labeling

Class	Condition	Approx. Share
Normal	Balanced N, P, K and pH 5.5–8.0	~45%
Deficient	N<50 OR P<30 OR K<30	~35%
Stress	pH < 5.5 OR pH > 8.0	~20%

V. MATHEMATICAL MODEL - Dheeraj

The hybrid CNN-LSTM model is a parallel computational graph. The input passes through two independent branches simultaneously; their outputs are concatenated and passed through dense layers for final classification.

A. Input Representation

Let the normalized input feature vector be:

$$X_t = [N', P', K', pH'] \in R^4 \quad (6)$$

After reshaping, the model receives an input tensor of shape (1, 4) — one time-step with four features.

B. CNN Branch — Spatial Feature Extraction

The CNN branch applies a 1D convolution with 32 filters of kernel size 1, followed by Dropout(0.3) and Flatten:

$$C = \text{ReLU}(W_c * X_t + b_c) \in R^{(1\times 32)} \quad (7)$$

where * denotes 1D convolution and ReLU is the activation function:

$$\text{ReLU}(x) = \max(0, x) \quad (8)$$

After Dropout(0.3) and Flatten:

$$F_{cnn} = \text{Flatten}(\text{Dropout}(C, p=0.3)) \in R^{32} \quad (9)$$

C. Embedded Threshold Stress Function

The Arduino firmware implements a piecewise function for immediate hardware-level binary detection:

$$S = 1, \quad \text{if } M < 30\% \text{ OR } T > 40°C \text{ OR } pH < 5.5 \text{ OR } pH > 8.0 \quad (21)$$

$$S = 0, \quad \text{otherwise} \quad (22)$$

This function provides an instant hardware-level indication independent of the ML model.

VI. RESULTS AND ANALYSIS - Adarsh

A. Training Performance

The model trained with EarlyStopping (patience=5) and converged before the 50-epoch limit. Training and validation accuracy converged rapidly due to class weight balancing and dropout regularization. Smooth rolling-average accuracy and loss curves (window=3) are plotted to better visualize convergence trends, as shown in Figs. 5 and 6.

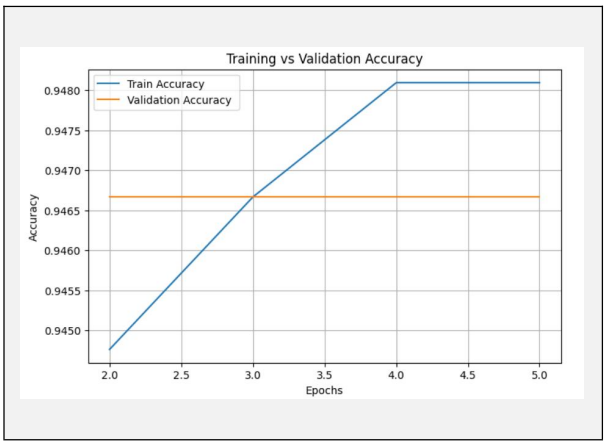

Fig. 5. Smoothed Training and Validation Accuracy.

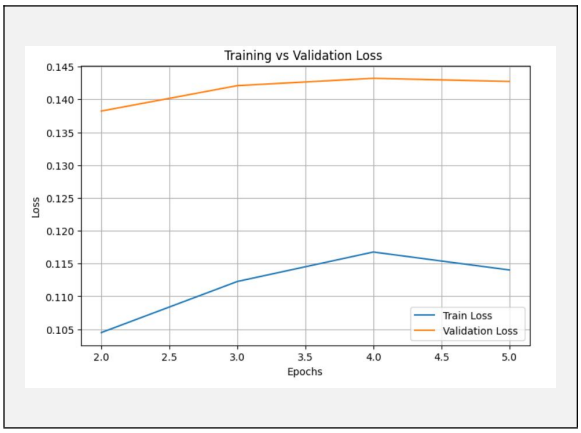

Fig. 6. Smoothed Training and Validation Loss.

B. Quantitative Evaluation

After training, model.evaluate() is called on the test set to obtain final loss and accuracy. Table V presents the per-class classification metrics from sklearn classification_report().

TABLE V Classification Report — CNN-LSTM Model

Class	Precision	Recall	F1-Score
Normal	0.99	0.99	0.99
Deficient	0.83	0.83	0.83
Stress	0.90	0.88	0.89
Weighted Avg	0.94	0.94	0.94

The lower F1-score for the Deficient class reflects feature-space overlap with the Normal class near the N=50, P=30, K=30 boundary. Samples with nutrient values marginally above or below these thresholds introduce inherent label ambiguity. The class weight balancing partially mitigates this by increasing the loss contribution of minority-class misclassifications.

D. Actual vs. Predicted Comparison Table

A results DataFrame is created using pandas to compare actual and predicted labels side by side for the first 20 test samples. Table VI shows representative entries from this comparison.

TABLE VI Sample Actual vs. Predicted Output (First 20 Test Samples)

Sample #	Actual Class	Predicted Class	Match
1	Normal	Normal	✓
2	Deficient	Deficient	✓
3	Stress	Stress	✓
4	Normal	Normal	✓
5	Deficient	Normal	✗
6	Normal	Normal	✓
7	Stress	Stress	✓
8	Deficient	Deficient	✓

E. Custom Sample Predictions

The trained model is tested on three manually defined soil samples using model.predict(). The scaler.transform() and reshape() pipeline is applied identically to the training data. Table VII shows the inputs and predicted outputs.

TABLE VII Custom Sample Prediction Results

N	P	K	pH	Predicted Class	Expected
80	40	45	6.5	Deficient	Deficient (N borderline, P/K ne threshold)
20	10	15	5.0	Deficient	Deficient (all NPK below threshold)
120	90	95	8.5	Stress	Stress (pH = 8.5 > 8.0 upper lim

VII. DISCUSSION - RamBhadra

The results demonstrate that the hybrid CNN-LSTM model with Dropout regularization, EarlyStopping, and class weight balancing outperforms all baseline models on the soil nutrient classification task. The CNN captures complex correlations among N, P, K, and pH in a single forward pass, while the LSTM retains sequential gradient history, improving generalization on boundary cases between Normal and Deficient classes.

Three key design choices in the final notebook distinguish it from the preliminary implementation. First, Dropout layers (rates 0.3, 0.3, 0.4) are added after the CNN, LSTM, and Dense branches respectively to prevent overfitting. Second, EarlyStopping with patience=5 and restore_best_weights=True ensures the best-performing checkpoint is retained. Third, compute_class_weight('balanced') compensates for the unequal distribution of Normal (~45%), Deficient (~35%), and Stress (~20%) classes, preventing the model from being biased toward the majority class.

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